

Autonomy without a signal: RhoA/Rho-kinase calcium sensitization as a mechanism for the persistence of hard flaccid syndrome

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Fifth in a series. Companion papers: *The sacral splanchnic bypass*; *Initiation without maintenance*; *The intraganglionic loop*; *A generator that outlives its cause*.

Abstract

Background. Every sympathetic account of hard flaccid syndrome (HFS) predicts that interrupting sympathetic outflow should abolish the sign. Sympathetic blockade is reported to fail. If those reports are accurate, the contraction is not being sustained by ongoing neural traffic at the time of treatment, and the lesion must lie in the effector tissue itself.

Hypothesis. We propose that cavernosal smooth muscle in HFS has become hypersensitive to calcium through up-regulation of RhoA/Rho-kinase signalling, so that it remains contracted at calcium concentrations that would normally leave it relaxed. Because the mechanism operates at the level of myosin light chain phosphatase rather than at calcium entry, the contraction persists when the neural signal is removed.

Basis. This is the best-established molecular pathway invoked anywhere in this series. RhoA/Rho-kinase-mediated calcium sensitization is one of the mechanisms maintaining normal penile flaccidity; Rho-kinase antagonism stimulates penile erection in the rat independently of nitric oxide; and Y-27632 antagonises noradrenergic contraction in *human corpus cavernosum*. What has never been measured is RhoA or Rho-kinase activity in tissue from a patient with HFS.

Two corrections to the conjecture as usually stated. First, the claim that α -blockade cannot reverse calcium sensitization because sensitization “operates downstream” treats calcium entry and calcium sensitization as parallel outputs. They are not. α 1-adrenoceptors couple through $G_{\alpha q}$ and $G_{\alpha 12/13}$ to RhoGEFs and thence to RhoA, so both arms are downstream of the same receptor, and Rho-kinase inhibition antagonises noradrenergic contraction in human cavernosal tissue. The hypothesis therefore survives only in a stronger form: the pathway must have become *receptor-independent*, driven by expression-level change rather than by ongoing agonist occupancy. Second, the proposed discriminating experiment — normalising resting tone at doses that do not themselves produce erection — may have no dose window, because Rho-kinase antagonism is itself erectogenic. We reformulate the prediction as a dose–response shift rather than a threshold effect.

Tests. No Rho-kinase inhibitor is approved for systemic use outside Japan and China, none is formulated for intracavernosal delivery, and cavernosal tissue is not obtainable in this population. We therefore propose a graded systemic challenge with a validated tone endpoint, a leukocyte Rho-kinase activity assay as an imperfect systemic surrogate, and — as the only test currently unobstructed — a retrospective analysis of whether α -blocker and phosphodiesterase-5 inhibitor response declines with symptom duration.

Conclusion. This conjecture supplies effector autonomy, which the reported block failures demand and no neural hypothesis provides. Its molecular foundation is solid and its clinical foundation is absent, and the experiment usually proposed to test it is harder to design than it appears.

Keywords: hard flaccid syndrome; RhoA; Rho-kinase; ROCK; calcium sensitization; myosin light chain phosphatase; corpus cavernosum; effector autonomy.

Reader advisory

This is hypothesis generation, not a report of new data. No component of the mechanism proposed here has been demonstrated in a person with hard flaccid syndrome. Rho-kinase inhibitors are discussed solely as research probes. None is approved for this indication anywhere; systemic Rho-kinase inhibition causes hypotension and other serious adverse effects; and intracavernosal administration of an unapproved compound carries a risk of priapism and tissue injury. Nothing here should influence clinical management or be used to seek any treatment.

1. Introduction

The four companion papers in this series all locate the lesion in the nervous system: in the route the sympathetic traffic takes [4], in the spinal circuits that generate it [5], in the ganglion where it meets the sensory system [6], or in the afferent axons themselves [7]. All four share a vulnerability. Each predicts that interrupting sympathetic outflow downstream should abolish the hard flaccid state, and sympathetic blockade at the superior hypogastric plexus is reported to fail [E] [3].

Those reports are unverified, undenominated, and unaccompanied by any documentation of technical adequacy; we have argued at length elsewhere that they cannot support a conclusion [4]. But they can support a question. *If* a technically confirmed sympathetic block genuinely fails to abolish the sign, then at the time of treatment the contraction is not being sustained by ongoing neural traffic. It is being sustained by the smooth muscle itself. The effector has become autonomous.

Several mechanisms could produce that autonomy [3]. This paper examines the one with the strongest molecular foundation: that the cavernosal smooth muscle has become hypersensitive to calcium through up-regulation of RhoA/Rho-kinase signalling, and therefore stays contracted at calcium concentrations that would normally permit relaxation.

The idea is attractive for a reason worth stating plainly. Almost every hypothesis about HFS invokes a plausible-sounding process that has never been characterised in erectile tissue. This one invokes a pathway that has been characterised in erectile tissue in considerable molecular detail, including in human samples — and then asks whether it is dysregulated in this condition. That is a smaller inferential leap than the others in the series require. It is also, as Section 4 shows, a leap that has been described somewhat inaccurately in the source material, and the correction matters.

2. Methods

This is a hypothesis and theory article. No new human or animal data were generated; no patients were involved; ethical approval was not required.

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *RhoA Rho-kinase erectile, calcium sensitization smooth muscle, Y-27632 corpus cavernosum, myosin light chain phosphatase, ROCK inhibitor clinical, G12/13 RhoGEF* and *hard flaccid syndrome*, with backward citation chasing. The search was purposive and non-systematic; no protocol was registered and no formal risk-of-bias appraisal was applied. We searched specifically for the receptor-coupling architecture of α 1-adrenoceptors, because the standard statement of this conjecture makes a claim about it, and Section 4 reports what that returned.

Claims are graded as in Table 1. Claims not verified against a retrievable source are marked [VERIFY]; quantities that do not exist in the literature are marked [DATA NEEDED].

Table 1. Evidence grading scheme. The grade refers to the strength of evidence for the *stated proposition*, not to its relevance to hard flaccid syndrome. No proposition here carries grade A evidence *in HFS*, because none exists.

Grade	Definition
[A]	Replicated human evidence: consistent findings across independent human studies, or human tissue pharmacology reproduced in more than one laboratory.
[B]	Single human study or human tissue study; human data with substantial methodological limitation.
[C]	Experimental animal or in vitro evidence, with species and preparation stated.
[D]	Mechanistic inference: deduced from A–C evidence but not itself directly observed.
[E]	Unverified anecdotal or community-sourced report; not peer reviewed, denominators unknown.

3. Background

3.1 Contraction is not governed by calcium alone

Smooth muscle force depends on the phosphorylation state of the 20 kDa myosin regulatory light chain, which is set by the balance between myosin light chain kinase — activated by calcium–calmodulin — and myosin light chain phosphatase (MLCP). Active RhoA, in its GTP-bound form, activates Rho-associated kinase (ROCK), which phosphorylates the myosin-targeting subunit MYPT1 and thereby inhibits MLCP. Light chain phosphorylation and force therefore increase without any change in intracellular calcium concentration; this is what is meant by calcium sensitization [A] [16, 17].

3.2 The pathway maintains normal penile flaccidity

This is not a marginal mechanism in erectile tissue. In the absence of arousal stimuli, Rho-kinase-mediated signalling promotes vasoconstriction of the cavernosal arterioles and sinuses, keeping the penis in the non-erect state [C] [9]. RhoA-mediated calcium sensitization has been characterised specifically in erectile function [C] [VERIFY] [10], and the role of RhoA/Rho-kinase in erectile tissue in health and disease has been reviewed at length [C] [VERIFY] [11].

Three experimental results anchor the claim:

1. **Rho-kinase antagonism produces erection, and does so without nitric oxide.** In the rat, intracavernosal Y-27632 raised corpus cavernosum pressure and initiated the erectile response independently of nitric oxide [C] [8]. Topical application to the tunica albuginea or glans produced the same effect, although high doses reduced systemic blood pressure [C] [VERIFY] [14].
2. **The pathway operates in human tissue.** Y-27632 antagonises noradrenergic contractions in rabbit and human penile corpus cavernosum [B] [VERIFY] [12], and human cavernosal smooth muscle cells express Rho-kinase [B] [VERIFY] [13].
3. **It interacts with the nitric oxide pathway.** Rho-kinase antagonism and phosphodiesterase-5 inhibition acted synergistically on the erectile response in hypertensive rats [C] [VERIFY] [15], and cyclic GMP-dependent protein kinase inhibits RhoA, providing a molecular basis for that interaction [C] [VERIFY: THE SPECIFIC RESIDUE AND ITS FUNCTIONAL SIGNIFICANCE IN CAVERNOSAL TISSUE WERE NOT CONFIRMED] [11].

3.3 What has never been done

No study has measured RhoA or Rho-kinase expression or activity in cavernosal tissue from a patient with hard flaccid syndrome, and no study has measured it in any patient with a condition characterised by pathologically *elevated* flaccid tone. The disease literature on this pathway in erectile tissue concerns states of impaired erection — diabetes, ageing, hypertension — in which elevated RhoA/ROCK activity is proposed to *oppose* erection [C] [VERIFY] [11]. Applying it to a disorder of excessive flaccid tone is an extrapolation, and one nobody has attempted [D].

We also identified no study demonstrating that sustained sympathetic or α -adrenergic stimulation up-regulates RhoA or Rho-kinase expression in cavernosal tissue [DATA NEEDED]. That step — the one that converts a neural cause into an effector-autonomous consequence — is the load-bearing assumption of this hypothesis and is currently unsupported in this tissue.

4. Two corrections to the conjecture as usually stated

The conjecture has been advanced with two supporting arguments that do not survive examination. Both can be repaired, but the repairs change what the hypothesis claims and how it should be tested. We set them out before stating the hypothesis, because the hypothesis should be stated in its corrected form.

4.1 Correction 1: calcium entry and calcium sensitization are not parallel

The argument usually offered is that α -blockade reduces adrenergically driven calcium entry but does not reverse calcium sensitization, which operates downstream at the level of myosin phosphatase — and that this is why α -blockade fails to normalise resting tone [3]. The picture behind that argument is of two parallel outputs, only one of which α -blockade reaches.

That picture is wrong. Pro-contractile G-protein-coupled receptors signal to RhoA principally through $G\alpha_{12/13}$, which binds and activates the RGS-homology-domain RhoGEFs — p115RhoGEF, LARG and PDZ-RhoGEF — and LARG can additionally be activated downstream of $G\alpha_q$ [A] [18, 19, 17]. α_1 -adrenoceptors are among the receptors that engage this architecture [B] [VERIFY] [20]. Calcium entry and calcium sensitization are therefore *in series* with the receptor, not in parallel with each other: both are consequences of agonist occupancy at the same receptor.

The functional confirmation is in the tissue at issue. If sensitization were independent of adrenergic signalling, Rho-kinase inhibition would not affect noradrenergic contraction. It does: Y-27632 antagonises noradrenergic contractions in human penile corpus cavernosum [B] [VERIFY] [12]. Rho-kinase is downstream of noradrenaline in this tissue, and blocking the receptor should therefore reduce Rho-kinase activation.

The repair. The hypothesis survives, but only in a stronger form: the pathway must have become *receptor-independent*. Sustained contraction must have produced a change at the level of expression or basal activity — more RhoA, more ROCK, more RhoGEF, or less MLCP — such that the tissue is sensitised even when adrenergic receptors are unoccupied. That is a different and more demanding claim than the one usually made, and it changes the required evidence: one would need to demonstrate elevated *basal* Rho-kinase activity in unstimulated tissue, not merely an exaggerated response to agonist [D].

4.2 Correction 2: the “sub-erectogenic dose” may not exist

The conjecture’s flagship prediction is that a Rho-kinase inhibitor should normalise resting tone *at doses that do not by themselves produce erection*, and that no other hypothesis predicts this [3].

The difficulty is that Rho-kinase antagonism is erectogenic. In the rat, intracavernosal Y-27632 raised corpus cavernosum pressure and initiated erection [8]; topical application did the same [14]. If reducing Rho-kinase activity in cavernosal tissue produces tumescence, then the dose that normalises pathologically elevated resting tone may lie at or above the dose that produces partial erection in a healthy tissue — in which case the proposed window is empty, and “normalised resting tone” cannot be distinguished from “partial erection” by any measure of tone.

The repair. Reformulate the prediction as a **dose–response shift rather than a threshold effect**. If HFS tissue is calcium-sensitised through Rho-kinase up-regulation, it should show *greater* tone reduction than control tissue at matched doses, and the relationship between dose and tone reduction should be shifted, not merely displaced vertically. That prediction survives whether or not a sub-erectogenic window exists, and it requires a control group — which the original formulation did not (Figure 1).

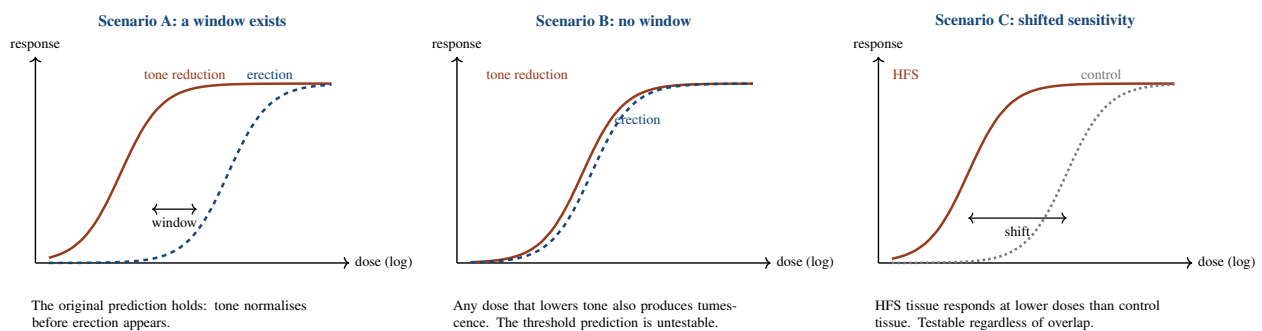


Figure 1. The dose-window problem, and the reformulated prediction. These are hypothetical scenarios drawn to illustrate an argument; they are not data, and no dose–response data exist for Rho-kinase inhibition in human cavernosal tissue *in vivo*. Scenario A is what the conjecture as originally stated assumes: a dose range in which resting tone normalises without producing erection. Scenario B is the possibility raised by the rodent finding that Rho-kinase antagonism is itself erectogenic [8, 14] — if the two curves coincide, “normalised resting tone” and “partial erection” are indistinguishable and the threshold prediction cannot be evaluated. Scenario C is the reformulation we propose: a leftward shift in sensitivity in hard flaccid syndrome relative to matched controls. It is testable whether or not a window exists, but it requires a control group, which the original formulation did not.

5. The hypothesis

Formal statement

H5 (RhoA/Rho-kinase calcium sensitization). In hard flaccid syndrome, cavernosal smooth muscle has undergone up-regulation of RhoA/Rho-kinase signalling, inhibiting myosin light chain phosphatase and raising the sensitivity of the contractile apparatus to whatever calcium is present. The tissue therefore remains contracted at calcium concentrations that would normally permit relaxation, and the contraction persists when neural drive is removed.

H5-autonomous (the form required by the block-failure reports). The sensitization is receptor-independent, sustained by expression-level or basal-activity change rather than by ongoing agonist occupancy. Prediction: basal Rho-kinase activity is elevated in unstimulated tissue; no adrenoceptor antagonist normalises resting tone; a Rho-kinase inhibitor does.

H5-driven (the weaker form). The sensitization is real but remains downstream of ongoing adrenergic drive. Prediction: adequate adrenoceptor blockade reduces both calcium entry and sensitization, and would normalise tone — meaning this form does *not* explain block failure and is not effector autonomy at all.

H5-null. Cavernosal calcium sensitivity in HFS is normal. Whatever maintains the state is neural, structural, or not present.

The distinction between the autonomous and driven forms is the whole substance of the hypothesis. **Only H5-autonomous does the explanatory work for which the conjecture was proposed**, and only H5-autonomous requires the receptor-independence established in Section 4. We recommend that the hypothesis be stated, tested and defended in that form, and that the weaker form be recognised as offering no advantage over a straightforwardly neural account.

Auxiliary assumptions:

- **A1 (induction).** Sustained sympathetic or α -adrenergic stimulation up-regulates RhoA/Rho-kinase in cavernosal tissue. Not demonstrated in this tissue [DATA NEEDED]. This is the hypothesis’s weakest link and it is upstream of everything else.

- **A2 (direction).** Elevated RhoA/Rho-kinase activity produces pathologically firm flaccidity rather than merely impaired erection. The existing disease literature associates elevated activity with erectile dysfunction [11]; that HFS patients also report erectile difficulty in 74.1% [B] [2] is consistent, but the primary sign here is excess flaccid tone and no disease model of that exists [DATA NEEDED].
- **A3 (persistence).** The up-regulation persists for years without continuing stimulus. Untested.

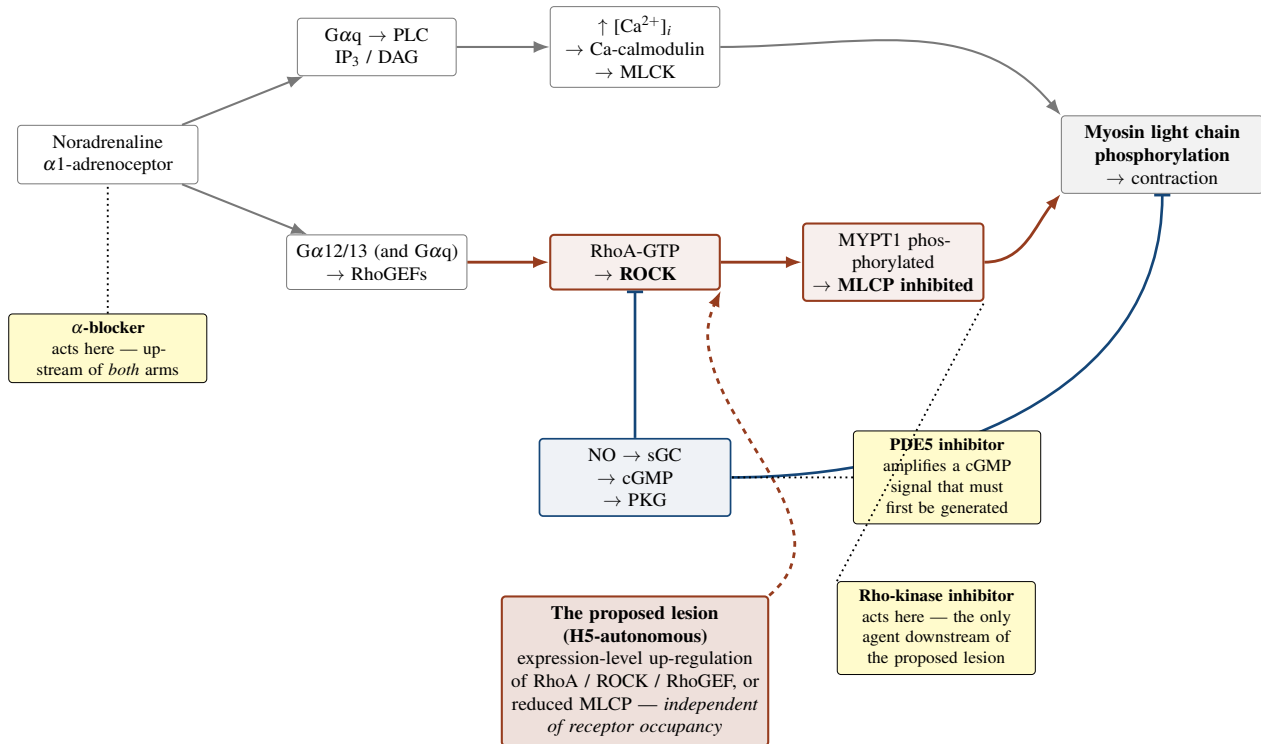


Figure 2. Signalling architecture of cavernosal contraction, and where each agent acts relative to the proposed lesion. The key correction to the conjecture as usually stated is visible here: calcium entry and calcium sensitization are not parallel outputs but two arms downstream of the same receptor, so an α -blocker reduces both. The hypothesis therefore requires the RhoA/ROCK arm to have become receptor-independent (dashed arrow, lower centre), driven by expression-level change rather than agonist occupancy. Nitric oxide opposes the pathway at two points, which is why a sufficiently large nitroergic drive can still produce erection in a sensitised tissue that never becomes normally soft — a quantitative rather than a qualitative argument. A Rho-kinase inhibitor is the only listed agent acting downstream of the proposed lesion. Diagram is schematic; the inhibition of RhoA by cGMP-dependent protein kinase is shown as reported in review, and the specific molecular details were not independently verified.

6. What the hypothesis explains

6.1 Effector autonomy

This is the reason the conjecture exists. If the reported block failures are accurate, the state is maintained without ongoing neural traffic, and no neural hypothesis in this series accounts for that. A tissue whose calcium sensitivity has been raised is contracted without needing a signal [D]. H5-autonomous supplies exactly the property the observation demands.

We restate the caveat: the observation is grade E, and the most parsimonious explanation for reported block failure remains technical inadequacy of the blocks [4]. H5 is the best available answer to a question that may not need answering.

6.2 Erection without normal softness

Patients describe a penis that can become erect but never becomes normally soft, and the disputed observation that phentolamine produces erection without normalising resting tone has been offered in support [3, 24]. Under H5 this is expected — but as a *quantitative* argument, not the qualitative one usually given.

Erection recruits a large nitroergic signal; the resting state does not. Nitric oxide opposes the RhoA/ROCK pathway as well as lowering calcium (Figure 2) [VERIFY] [11, 15]. A sufficiently large cGMP surge can therefore overcome a sensitised tissue and produce erection, while the much smaller tonic nitroergic activity present at rest cannot. The

observation is explained by the size of the opposing signal, not by the sensitization being unreachable [D]. We flag this because the qualitative version of the argument — that erection uses “a different and more powerful relaxation mechanism” immune to the lesion — is not accurate as stated.

6.3 Mixed response to phosphodiesterase-5 inhibitors

If the tissue is sensitised, an agent that amplifies an existing cGMP signal will work better where nitrgergic tone is better preserved. Mixed response across patients becomes a function of the size of the endogenous signal rather than of constitutional differences [D]. This is a genuine explanatory gain, but it is shared with at least two other conjectures — impaired nitric oxide generation, and fibrotic replacement of smooth muscle [3] — and does not discriminate between them.

7. Feasibility: the obstacles are severe

The conjecture is usually presented with an accompanying experiment: give a Rho-kinase inhibitor and see whether resting tone normalises. Four obstacles stand between that sentence and a study.

1. **No suitable drug is available in most jurisdictions.** Two Rho-kinase inhibitors are approved for clinical use in Japan (fasudil and ripasudil) and one in China (fasudil); ripasudil (2014, Japan) and netarsudil (2017, United States) are ophthalmic preparations for glaucoma and ocular hypertension [B] [VERIFY] [21, 22]. Fasudil, the only systemically administered agent among them, is approved for cerebral vasospasm and is not licensed in the United States or the European Union [VERIFY]. Newer orally available and isoform-selective agents exist for other indications, and their availability should be checked at the time any study is designed [VERIFY: WE DID NOT CONFIRM THE CURRENT REGULATORY STATUS OF ORAL ROCK2-SELECTIVE AGENTS AND READERS SHOULD NOT RELY ON OUR ACCOUNT OF IT].
2. **Systemic Rho-kinase inhibition is not benign.** Reported fasudil toxicity includes hypotension, nausea, haemorrhage, pyrexia and renal failure, and long-term use is described as causing severe complications [B] [VERIFY] [23]. Systemic exposure sufficient to affect cavernosal tone in a young man with a non-life-threatening condition would require substantial ethical justification.
3. **No intracavernosal formulation exists.** The rodent work used intracavernosal or topical Y-27632, a research compound [8, 14]. Intracavernosal administration of an unapproved agent risks priapism and tissue injury and should not be contemplated outside a formal first-in-human programme.
4. **The tissue is unobtainable.** Measuring RhoA or Rho-kinase activity in cavernosal tissue requires corporal biopsy. There is no clinical indication for corporal biopsy in HFS, and the population is young and otherwise well. The direct molecular test of this hypothesis is, for practical purposes, unavailable [DATA NEEDED].

The honest summary is that **the best-founded molecular hypothesis in this series is also the hardest to test in the population that matters.** That asymmetry is worth stating explicitly, because it is easy to mistake the strength of a mechanism’s basic-science foundation for the strength of the case that it operates in a given disease.

8. Predictions and tests

8.1 The test that is currently unobstructed

Prediction P1. If calcium sensitization accumulates with disease duration, then response to α -adrenoceptor blockade and to phosphodiesterase-5 inhibitors should decline as symptom duration increases.

This requires no drug, no procedure and no new instrument. It is checkable retrospectively from existing clinical records and from re-analysis of existing survey data, provided duration was recorded. It is weak evidence — confounded by treatment sequencing, by attrition of responders from clinical follow-up, and by recall — but it costs almost nothing and no one has done it. Notably, the same prediction follows from at least two competing conjectures [3], so a positive result would support a class of effector hypotheses rather than H5 specifically.

8.2 Systemic surrogate: leukocyte Rho-kinase activity

Because cavernosal tissue is unobtainable, a systemic surrogate is worth considering. Rho-kinase activity has been assessed in circulating leukocytes as a biomarker in cardiovascular research, typically by measuring phosphorylation of MYPT1 or of the ROCK substrate ezrin/radixin/moesin [B] [VERIFY: WE DID NOT CONFIRM THE ASSAY’S VALIDATION STATUS OR ITS REFERENCE RANGES, AND READERS SHOULD VERIFY BEFORE DESIGNING AROUND IT].

Prediction P2. Leukocyte Rho-kinase activity is elevated in men with HFS relative to age-matched controls.

We are not enthusiastic about this test and state why. A hypothesis about a local, tissue-specific, injury-induced change in cavernosal smooth muscle does not obviously predict a systemic biomarker abnormality, and a negative result would therefore not refute it. A *positive* result would be more informative than a negative one, and would raise the further question of whether HFS involves a systemic rather than local process — which would be a substantial finding in its own right.

8.3 Graded systemic challenge, if a suitable agent is available

Prediction P3 (reformulated from the conjecture’s original threshold prediction; see Section 4.2). Under a graded systemic Rho-kinase inhibitor challenge, men with HFS show a leftward-shifted dose–response for reduction in objectively measured resting cavernosal tone compared with age-matched controls.

Design requirements: a control group, which the original formulation omitted; an objective tone endpoint, which does not yet exist [4, 5]; blinded outcome assessment; blood-pressure monitoring as a safety endpoint and as an internal control for systemic drug effect; and pre-specified handling of the possibility that erection supervenes. Given Section 7, we regard this as a study to design after a tone endpoint exists and after a suitable agent has been identified, not before.

8.4 The decisive test, and why it is unavailable

Prediction P4. RhoA and Rho-kinase expression and basal activity are elevated in cavernosal tissue from men with HFS relative to controls, and basal MYPT1 phosphorylation is increased in unstimulated tissue.

The requirement for *basal* rather than agonist-evoked measurement follows from Correction 1: an exaggerated response to noradrenaline would be compatible with the driven form of the hypothesis, which explains nothing that a neural account does not. Only elevated basal activity demonstrates autonomy.

This is the experiment that would settle the question, and it requires tissue that cannot ethically be obtained in this population. We note it for completeness and to mark the boundary of what is currently knowable [DATA NEEDED].

Table 2. Predictions of H5 compared with the principal alternatives.

Observation	H5 (calcium sensitization)	Neural drive (any route)	Fibrotic remodelling	Discriminating value
Confirmed sympathetic block fails to reduce tone	Predicted	Not predicted	Predicted	Moderate–high
α -blockade fails to normalise resting tone	Predicted (autonomous form only)	Not predicted	Predicted	Moderate
Rho-kinase inhibitor reduces resting tone more in HFS than controls	Predicted	Not predicted	Not predicted	High
Elevated <i>basal</i> Rho-kinase activity in unstimulated cavernosal tissue	Predicted	Not predicted	Not predicted	Decisive but unobtainable
Erection achievable but resting tone never normalises	Predicted	Not predicted	Predicted	Low
Response to PDE5 inhibitors declines with duration	Predicted	Not predicted	Predicted	Low
Reduced smooth muscle content or increased collagen on imaging	Not predicted	Not predicted	Predicted	High, against H5

9. Alternative explanations

1. **Technical failure of the blocks.** The observation that motivates the entire effector-autonomy class is grade E and undocumented [3, 4]. If the blocks were inadequate, there is no autonomy to explain and H5 loses its rationale entirely.
2. **Other routes to autonomy.** Denervation supersensitivity, co-transmitter dominance and pacemaker dysregulation in interstitial cells would each produce effector autonomy by different means [3]. They are not mutually exclusive with H5 and are not distinguished from it by any test proposed above except the tissue assay.
3. **Fibrotic remodelling.** Replacement of smooth muscle by collagen would reduce compliance and produce a firm, poorly distensible organ without any change in calcium sensitivity [3]. **This is H5’s most serious competitor**, because it explains the same clinical picture, predicts the same declining drug response with duration, and — unlike H5 — is testable now by ultrasound elastography or magnetic resonance imaging. Any study of H5 should include a structural imaging arm, because a positive structural finding would substantially weaken the case for a purely functional mechanism.
4. **Wrong direction of effect.** The disease literature on RhoA/ROCK in erectile tissue concerns impaired erection, not

excessive flaccid tone [11]. It is possible that elevated Rho-kinase activity in HFS, if present, is a consequence of chronic hypoxia or reduced perfusion rather than a cause of the tone abnormality **[D]**.

5. **The premise may be false.** As in every paper in this series: cavernosal tone has never been objectively shown to be elevated in HFS, and sympathetic outflow to the penis has never been measured in an affected person.

10. Implications

No clinical implications are drawn, and we specifically caution against inferring any from the availability of Rho-kinase inhibitors for other indications. Three research implications follow.

First, state the hypothesis in its autonomous form or not at all. The driven form explains nothing that a neural account does not, and conflating the two makes the hypothesis look stronger than it is.

Second, include a structural arm. Fibrotic remodelling predicts almost everything H5 predicts and is testable with existing imaging. Any study designed to test H5 that omits structural imaging cannot interpret its own negative results.

Third, do the free study. Whether α -blocker and phosphodiesterase-5 inhibitor response declines with symptom duration is checkable from records that already exist. It is weak evidence, but it is available now, and the whole effector-autonomy class of hypotheses rests on an assumption about disease staging that nobody has examined.

11. Limitations

1. **No evidence in HFS.** RhoA and Rho-kinase have never been measured in tissue from a patient with this condition, or with any condition of pathologically elevated flaccid tone.
2. **The induction step is unsupported.** We identified no evidence that sustained adrenergic stimulation up-regulates RhoA/Rho-kinase in cavernosal tissue. This assumption is upstream of the entire hypothesis.
3. **The direction of the disease association is opposite.** Elevated RhoA/ROCK in erectile tissue is described in states of impaired erection, not excessive flaccidity.
4. **Two of the conjecture's supporting arguments required correction** (Section 4), and the corrections make the hypothesis more demanding rather than less.
5. **The flagship experiment may have no dose window**, and the reformulated version requires both a control group and a tone endpoint that does not exist.
6. **Feasibility is poor on four independent grounds** (Section 7): drug availability, systemic toxicity, absence of a local formulation, and inaccessibility of tissue.
7. **The motivating observation is grade E.** The entire effector-autonomy class rests on unverified reports of block failure in which adequacy was never documented.
8. **H5 is not distinguished from its sibling conjectures** by any currently available test, and is not distinguished from fibrotic remodelling by any test except imaging.
9. **Several citations, including three that are load-bearing, are flagged as unverified**, having been traced only through secondary reference lists. The human tissue pharmacology in particular [12] should be confirmed in full before this argument is relied on.
10. **Confirmation risk.** This paper was written to develop a conjecture that its author regards as the strongest of the set. We have tried to counter that by correcting two of its supporting arguments against it, by identifying the dose-window problem, and by naming fibrotic remodelling as a competitor that predicts nearly everything H5 predicts and is easier to test. The reader should still discount accordingly.

12. Refutation criteria

- **Refuted.** RhoA and Rho-kinase expression and basal activity in cavernosal tissue from men with HFS are normal against matched controls.
- **Substantially refuted.** A graded Rho-kinase inhibitor challenge shows no leftward shift in the dose–response for tone reduction in HFS relative to matched controls.
- **Substantially weakened.** A technically confirmed sympathetic block reliably abolishes objectively measured resting tone, removing the effector-autonomy problem the hypothesis exists to solve.

- **Substantially weakened.** Adequate α -adrenoceptor blockade normalises resting tone, indicating that any sensitization present remains receptor-dependent.
- **Weakened.** Structural imaging demonstrates reduced smooth muscle content or increased collagen proportional to symptom duration, favouring a structural over a functional mechanism.
- **Rendered moot.** Direct measurement shows resting cavernosal tone is not objectively elevated in symptomatic patients.

13. Conclusion

This conjecture has the best molecular foundation of any in the series and the worst experimental access. RhoA/Rho-kinase-mediated calcium sensitization is not a speculative pathway: it is characterised in erectile tissue, demonstrated in human corpus cavernosum, and known to be one of the mechanisms that maintains ordinary flaccidity. Proposing that it is turned up in a disorder of excessive flaccid tone is a modest extrapolation, and it supplies the one property the reported treatment failures demand and no neural hypothesis provides — a tissue that stays contracted without being told to.

Two of the arguments usually offered for it do not hold. Calcium entry and calcium sensitization are not parallel outputs that an α -blocker can reach only one of; they are two arms downstream of the same receptor, and Rho-kinase inhibition antagonises noradrenergic contraction in human cavernosal tissue. The hypothesis therefore requires the pathway to have become receptor-independent, which is a stronger claim demanding evidence of elevated *basal* activity rather than exaggerated agonist response. And the experiment usually proposed — normalising tone at sub-erectogenic doses — may have no dose window at all, because Rho-kinase antagonism is itself erectogenic. Reformulated as a dose–response shift against matched controls, the prediction survives; as a threshold effect, it may be untestable.

What remains is a hypothesis that is probably right about the pathway's importance in this tissue, entirely unevidenced about its state in this disease, and blocked from its decisive test by the unavailability of cavernosal tissue in young men who have no reason to undergo biopsy. Two things can be done anyway. Structural imaging can be added to any study, because fibrotic remodelling predicts nearly everything this hypothesis predicts and is testable today. And someone can check whether drug response declines with symptom duration, using records that already exist. Neither will confirm the hypothesis. Both would tell us whether it is worth the effort of trying.

Declarations

Authorship and the use of artificial intelligence. The second listed author is a large language model. It is listed at the corresponding author's request and in the interest of transparency, but this contravenes the recommendations of the International Committee of Medical Journal Editors, which hold that artificial intelligence tools cannot meet authorship criteria because they cannot take responsibility for the accuracy and integrity of the work. **Before submission to any peer-reviewed venue this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** Every citation, and particularly those marked [VERIFY], should be checked against the primary source first. The regulatory-status statements in Section 7 are time-sensitive and must be re-checked before submission.

Ethics approval. Not applicable. No human participants, animals or identifiable data were involved.

Funding. None.

Competing interests. No competing financial interests. The first author maintains a public website on hard flaccid syndrome and authored the source article [3], in which this conjecture is described as the strongest of those proposed; this is a non-financial interest in the hypothesis being taken seriously, and readers should weigh the argument accordingly.

Data availability. No datasets were generated or analysed.

Clinical disclaimer. This article is hypothesis generation. It does not constitute medical advice and does not establish a clinician–patient relationship. Nothing in it should be used to select, request or avoid any treatment. Rho-kinase inhibitors are not approved for this or any urological indication; systemic administration causes hypotension and other serious adverse effects; and intracavernosal injection must never be self-administered. Seek emergency care for an erection lasting more than four hours.

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